Results: Citation Network

RQ4: Citation Geometry I

Resolving each record's references against the corpus yields an intra-corpus citation graph (built and analyzed with NetworkX [@hagberg2008exploring]) of 2234 nodes and 8,623 edges across 1409 connected components, with a graph density of 0.17% and a mean in-degree of 3.9. Of 38,489 total outgoing references, 22.4% resolve to another record inside the corpus — a resolution rate that reflects how self-contained the retrieved slice of the literature is rather than the underlying citation density of any single work.

The citation network has 1416 communities (detected by modularity optimization), a maximum in-degree of 163 (the most-cited paper within the corpus), and a maximum out-degree of 143 (the paper that cites the most other corpus members).

Centrality Analysis I

Centrality scores (PageRank [@page1999pagerank] and HITS) and modularity-based community detection [@clauset2004finding] are rounded and ranked with a stable tiebreaker so the reported hub and authority rankings are byte-reproducible across runs despite the floating-point non-associativity of the underlying iterative solvers.

Table 4. Top 5 papers by PageRank.

Rank	DOI	PageRank
1	10.1177/026988110001400107	0.036943
2	10.1212/wnl.54.5.1166	0.026949
3	10.1523/jneurosci.21-05-01787.2001	0.013668
4	10.1523/jneurosci.20-22-08620.2000	0.011349
5	10.4088/jcp.v61n0510	0.008136

Centrality Analysis II

Table 5. Top 5 authority papers (HITS).

Rank	DOI	Authority
1	10.1038/sj.npp.1301534	0.017700
2	10.1007/s00213-002-1250-8	0.016311
3	10.1124/jpet.106.106583	0.015144
4	10.1523/jneurosci.21-05-01787.2001	0.014734
5	10.1001/jama.2009.351	0.014481

Centrality Analysis III

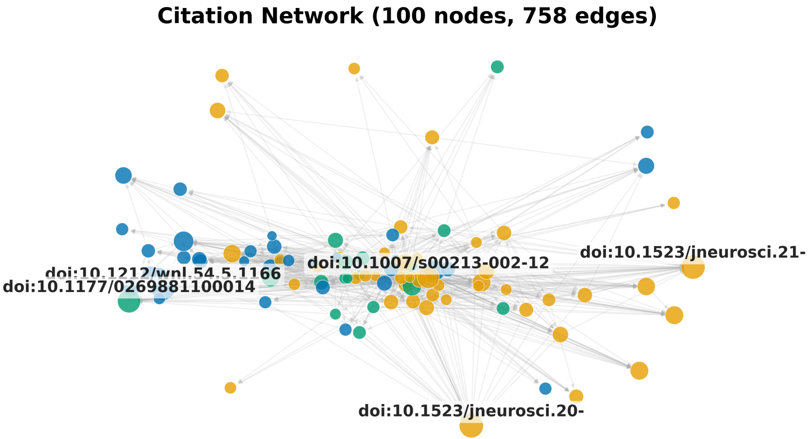
Table 6. Top 5 hub papers (HITS).

Rank	DOI	Hub
1	10.3389/fnins.2021.656475	0.011968
2	10.1016/bs.apha.2023.10.006	0.011596
3	10.1038/sj.npp.1301534	0.010886
4	10.1080/08897077.2019.1700584	0.010562
5	10.1007/s00213-013-3232-4	0.009896

Centrality Analysis IV

The most influential paper by PageRank (DOI 10.1177/026988110001400107) is a foundational work that anchors the citation structure — its high authority score confirms it is frequently cited by other corpus members. Hub papers, which cite many other corpus members, serve as integrative reviews or meta-analyses that connect disparate threads of the literature.

Centrality Analysis V



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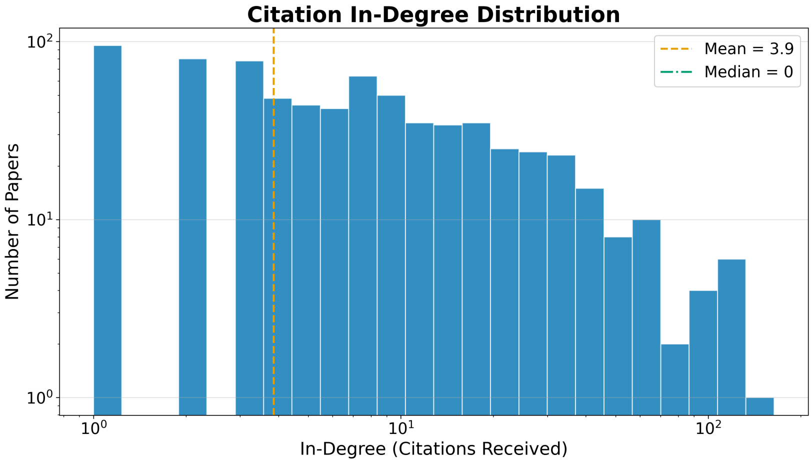
Centrality Analysis VI



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Centrality Analysis VII

The heavy-tailed degree distribution is characteristic of citation networks: a small number of highly-cited papers anchor the structure, while the long tail of low-degree nodes represents newer or peripheral works. The low graph density (0.17%) reflects the sparsity of intra-corpus citation links — most papers cite works outside the retrieved slice, which is expected for a max-results-capped retrieval.

Advanced Network Metrics I

Beyond PageRank and HITS, the network analysis computes betweenness centrality (which papers bridge different communities), closeness centrality (which papers are near all others), degree assortativity (do highly-cited papers cite other highly-cited papers?), and average clustering coefficient (how tightly knit are local neighborhoods).

The degree assortativity coefficient is -0.0598, and the average clustering coefficient is 0.1029. A negative assortativity indicates that highly-cited papers tend to cite less-cited papers (dissortative mixing), which is typical of citation networks where review papers (high in-degree) cite many primary studies (low in-degree).

Table 7. Top 5 papers by betweenness centrality.

Rank	DOI	Betweenness
1	10.1038/sj.npp.1301534	0.005234
2	10.4088/jcp.v67n0406	0.003451
3	10.1016/j.neuropharm.2012.07.011	0.002530
4	10.1002/14651858.cd006788.pub3	0.002289

Advanced Network Metrics II

5	10.2165/00003495-200868130-00003	0.001894

Advanced Network Metrics III

Papers with high betweenness centrality serve as bridges between different topical communities in the citation network — their removal would fragment the graph into disconnected components. These bridging papers are often review articles or methodological papers that connect disparate research threads.